

WORKSHEET A.5.M2

Network Operation Check

What we will learn today:

- Display useful network information.
- Use IP addresses.
- Identify problems in small networks and propose and apply solutions.
- Check network speed.

Consolidation / Assessment Activities

1. For each IPv4 address and subnet mask assigned to Computer 1, decide whether the IPv4 address of Computer 2 belongs to the same network.

No.	IPv4 - Computer 1	Subnet Mask	IPv4 - Computer 2	Yes/No
1	192.168.10.47	255.255.255.0	192.168.20.47	
2	192.168.1.15	255.255.255.0	192.168.1.200	
3	192.168.10.47	255.255.0.0	192.168.20.47	
4	10.1.2.3	255.0.0.0	10.3.2.1	
5	172.16.32.25	255.255.0.0	172.17.40.68	

2. For cases 1-5 above, state where using the Default Gateway is compulsory so that Computers 1 and 2 can communicate.

The following commands use the Windows Command Prompt. To open it, select Search, type Command, and choose Command Prompt.

3. Type ipconfig at the command prompt and press Enter. Record your computer's network information:

- (a) IPv4 Address: ____ . ____ . ____ . ____
(b) Subnet Mask: ____ . ____ . ____ . ____
(c) Default Gateway: ____ . ____ . ____ . ____

4. Type `ipconfig /all` at the command prompt and press Enter. Record the following additional information:

(a) MAC/Physical Address: ____-____-____-____-____-____

(b) DNS Servers: ____ . ____ . ____ . ____
____ . ____ . ____ . ____

5. Type `nslookup` at the command prompt and press Enter.

(a) Enter `www.moec.gov.cy` and record its IP address: ____ . ____ . ____ . ____

(b) Enter `google.com` and record its IP address: ____ . ____ . ____ . ____

To end `nslookup`, type `exit`.

6. What is the role of a DNS server?

(a) We send it an IP address and it returns a computer name.

(b) We send it a computer name and it returns its IP address.

(c) We send it an IP address and it returns the country where the computer is located.

(d) We send it a computer name and it returns its MAC/Physical address.

7. Open a web browser and enter `142.250.201.46` in the address bar. Which website appears? _____

8. At the command prompt, type `ping 142.250.201.46` and press Enter. What does this command test?

(a) Whether our computer is connected to the Internet.

(b) Whether our computer can communicate with the computer at `142.250.201.46`.

(c) Whether both computers belong to the same network.

(d) Whether our computer is connected to the local network.

9. Type `tracert 142.250.201.46` and press Enter. How many routers (hops) are used before reaching the destination? _____

10. Repeat using `tracert www.moec.gov.cy`. How many routers (hops) are used?

11. Visit www.speedtest.net and select GO.

(a) Download speed: _____

(b) Upload speed: _____

12. Match each check/problem in Column A with the correct action in Column B. Write the number from Column B beside each item in Column A.

Column A - Check/Problem	Column B - Action
(a) Check communication between the computer and router.	1. Restart the computer and/or router.
(b) Check whether the computer sends/receives data through the cable.	2. Check the indicator light on the network card.
(c) Check whether the computer received a valid address from the router.	3. Use the Network Troubleshooter.
(d) Use Windows to diagnose and repair the problem.	4. Use the ipconfig command.
(e) The computer has not received an IP address from the router.	5. Ping the Default Gateway IP address.

Additional Activities

13. Complete the missing part of Computer 2's IPv4 address so that it does or does not belong to the same network as Computer 1.

No.	IPv4 - Computer 1	Subnet Mask	IPv4 - Computer 2	Same?
1	192.168.25.72	255.255.255.0	192.168.___.65	Yes
2	192.168.39.150	255.255.255.0	192.168.___.20	No
3	192.168.202.3	255.255.0.0	192.168.___.7	Yes
4	10.10.10.10	255.0.0.0	10.___.11.11	Yes
5	172.16.30.2	255.255.0.0	172.___.30.6	No

Activities for Home

14. Record the following network information for your computer:

(a) IPv4 Address: ____ . ____ . ____ . ____

(b) Subnet Mask: ____ . ____ . ____ . ____

(c) Default Gateway: ____ . ____ . ____ . ____